

GCSE Maths Formulae (2026 exams)

Formulae provided within exam questions

The following formulae are provided directly within the body of exam questions when they are needed. Students do not need to memorise these formulae, but they do need to know how to use them.

Curved surface area of a cone	$\pi r l$
Surface area of a sphere	$4\pi r^2$
Volume of a cone	$\frac{1}{3}\pi r^2 h$
Volume of a sphere	$\frac{4}{3}\pi r^3$
Volume of a pyramid	$\frac{1}{3} \times \text{area of base} \times \text{perpendicular height}$

In addition to these formulae provided within questions, students also receive a separate [formulae sheet](#) with all their GCSE maths exams containing many other formulae.

Formulae NOT provided - Must be memorised

Area of a rectangle	length x width
Area of a parallelogram	base x perpendicular height
Area of a triangle	$\frac{1}{2} \times \text{base} \times \text{perpendicular height}$
Volume of a cuboid	length x width x height
Volume of a cylinder	$\pi r^2 h$
Speed, distance and time	$\text{speed} = \frac{\text{distance}}{\text{time}}$
Density, mass and volume	$\text{density} = \frac{\text{mass}}{\text{volume}}$
Sum of interior angles in a polygon	$(n - 2) \times 180$
Gradient	$\frac{\text{change in } y}{\text{change in } x}$